

Solution to “Relish” the Challenge

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Preface

Provided is just one of the solutions that you can use. The solution has been broken up into parts based on significant breakthroughs in the puzzle. At the top of each page after this one will be the significant breakthrough(s), followed by the explanation. Sometimes other numbers that can be discovered using this information will also be revealed. There will also be an image of the current board after all these other numbers are inserted into the grid. This way, you can look at the top of each page to get only the next hint.

In the solution, we will define the grid to be the entire 9 x 9 field. Each 1 x 1 subgrid will be called a cell. A cell is denoted by a column, given a letter from A to I, and a row, given a number from 1 to 9. For example, E5 will denote the center cell of the grid. A cell is *solved* when the number that goes inside that cell is limited to one value. A clue represents an arithmetic rule for one or more cells that describes the relationship of the numbers in these cells. A group is a group of cells that share the same clue. Groups are bounded by bold lines in the grid. A cell is said to be *occupied* by a digit when that digit appears in that cell in the final solution. Standard KenKen rules apply.

	A	B	C	D	E	F	G	H	I
1	?			21+	2/		?		12x
2	4536x		42x			84x		?	
3					36x				24x
4			28x				14+		
5	?	1-		6-	?				17+
6			3/		?				
7		?		36x	36x		?		3/
8								?	
9		?							

Step 1: Cell C5 is a 7

Explanation:

In Rows 2 through 4 are many clues that involve multiplication to a value that is a multiple of 7. Because 7 is prime, a 7 must occupy at least one cell in each of these groups. In Rows 2 and

3 specifically, there is a 42x clue and a 84x clue. These two groups will each require a 7. Because each row can only have one 7, the 7s in Rows 2 and 3 will be used in these two groups.

The prime factorization of 4536 is $2^3 3^4 7$. Since 7 divides 4536, 7 must occupy one of the cells in this group. We just claimed that the 7s in Rows 2 and 3 are used elsewhere, so the 7 in the 4536x group cannot appear in Row 2 or 3. Thus, it must appear in Row 4.

We also see that 7 divides 28. Since the 7 in Row 4 occupies either cell A4 or cell B4, the 7 required in the 28x group cannot appear in Row 4. Then the 7 must occupy the last cell in this group: cell C5.

Follow-up:

Returning to the 42x group, with a 7 occupying cell C5, the 7 in the 42x group cannot be in Column C. Then 7 must occupy cell B3. Likewise, the 7 from the 4536x clue that was originally restricted to Row 4 can no longer occupy cell B4. Therefore, it must occupy cell A4.

	A	B	C	D	E	F	G	H	I
1	?			21+	2/		?		12x
2	4536x		42x			84x		?	
3		7			36x				24x
4	7		28x				14+		
5	?	1-	7	6-	?				17+
6			3/		?				
7		?		36x	36x		?		3/
8								?	
9		?							

Step 2: Cell I3 is a 1

Explanation:

In Row I there are many clues that can be solved with the same numbers. In particular, these are the 12x, 24x, and 3/ clues. 12x can be solved by (2,6) or (3,4). The 3/ can be solved by (1,3), (2,6), or (3,9). Notice that by the rule that each number must appear exactly once in each column, then the 2, 3, and 6 are all used within these two groups. Imagine that the 12x was solved by (2,6). Then the 3/ cannot be solved by the (2,6). Both (1,3) and (3,9) use a 3, so in the end, 2, 3, and 6 were all used. Similarly, if the 12x was solved by (3,4), then the 3 cannot appear in the 3/ clue, so 3/ must be solved by (2,6). With this in mind, we can force the options for the 24x clue.

Because the 24x clue spans two columns and two rows, we can allow one digit to be repeated once. This gives four options to fill the 24x group: (1,3,8), (1,4,6), (2,2,6), and (2,3,4). Notice that since we already used the 2, 3, and 6 in Column I, we can only have one copy of 2, 3, or 6 in the 24x group because only one cell exists outside of Column I. Then we can eliminate the last two.

Out of the remaining two options (1,3,8) and (1,4,6), a 1 will be present in the 24x clue. Let's return to the 28x clue. 28x can be solved in three cells with (1,4,7) and (2,2,7). Since the 7 solves cell C5, the remaining two digits cannot be the same, because they appear in the same row.

Therefore, 28x must be solved by (1,4,7). Since a 1 is confirmed to appear in either cell C4 or cell D4, then there cannot be any more 1s in Row 4. Therefore, the 1 in the 24x clue must appear in cell I3.

	A	B	C	D	E	F	G	H	I
1	?			21+	2/		?		12x
2	4536x		42x			84x		?	
3		7			36x				1
4	7		28x				14+	24x	
5	?	1-	7	6-	?				17+
6			3/		?				
7		?		36x	36x		?		3/
8								?	
9		?							

Step 3: Cell E6 is a 7

Explanation:

There are only so many places any particular digit can be in any row or column. In Column E, 7 cannot be in the 2/ group in Row 1 or the 36x group in Rows 7 through 9. From the 7s we have already deduced to solve cells A4, B3, and C5, there cannot be any more 7s in this row. Therefore, the 7 in Column E cannot appear in cells E3, E4, or E5. In the 84x clue that spans Rows 2 and 3, we need a 7 in this group. There cannot be a 7 in Row 3, so it must appear in Row 2. Then there cannot be any more 7s in this row. In particular, cell E2 is not a 7. This leaves one spot for the 7 in Column E: cell E6.

Follow-up:

We can use similar reasoning with the 7 in Column I to deduce that cell I9 is also solved by a 7. Then cell D1 is a 7. We have thus placed six 7s.

Step 4: Cell B4 is a 9

Explanation:

Consider the 36x group in Column E. All three digits must be distinct because the three cells appear in the same column. There are two ways to solve this clue: (1,4,9) and (2,3,6). Now consider the 36x group in Columns E and F. With the repeat, there are five options to solve this clue: (1,4,9), (1,6,6), (2,2,9), (2,3,6), and (3,3,4).

Let's eliminate some of these possibilities. The 28x group must be solved by a (1,4,7) and not a (2,2,7) because the two empty cells are in the same row. Then there cannot be any other 1s in Row 4. The 1 in Row 3 solves cell I3. Then there cannot be a 1 in the 36x group in the middle. It must be solved by (2,2,9), (2,3,6), or (3,3,4).

Returning to the vertical 36x group in Column E, assume it is solved by (2,3,6). If this were the case, none of the three remaining options for the 36x above it could solve it because there is only one spot for a 2, 3, or 6. For example, if the vertical 36x group was (2,3,6) then the 36x group

	A	B	C	D	E	F	G	H	I
1	?			7	2/		?		12x
2	4536x		42x	21+		84x		?	
3		7			36x				1
4	7		28x				14+	24x	
5	?	1-	7	6-	?				17+
6			3/		7				
7		?		36x	36x		?		3/
8								?	
9		?							7

above it cannot be solved by (2,2,9) because the 2 in Column E will be placed in the vertical group, and the two 2s in the middle 36x group cannot both be in cell F4.

Since none of the remaining possible solutions for the middle 36x group will work with the bottom being solved by (2,3,6), the vertical 36x clue must be solved by (1,4,9). With this in mind, the middle 36x group cannot be solved by (1,4,9). It also cannot be solved by (2,2,9), (3,3,4), or (1,6,6) because this would force the 9 or 1 to be in cell E4. Then the middle 36x group must be solved by (2,3,6).

In Row 4, the 9 cannot be in the 28x or 24x clues because of divisibility. The middle 36x clue is solved by (2,3,6) so the 9 cannot solve cells E4 or F4 either. If cell G4 was solved by a 9, then the remaining four cells of the 14+ group must add to 5, which is impossible given its shape. Therefore, the 9 must solve the last remaining cell in Row 4: cell B4.

Follow-up:

Divisibility also shows that cell G4 must be solved by a 5.

	A	B	C	D	E	F	G	H	I
1	?			7	2/		?		12x
2	4536x		42x	21+		84x		?	
3		7			36x				1
4	7	9	28x				5	24x	
5	?	1-	7	6-	?	14+			17+
6			3/		7				
7		?		36x	36x		?		3/
8								?	
9		?							7

Step 5: Cell I4 is an 8

Explanation:

In Row 4, we have solved cells A4, B4, and G4. We know that cells C4 and D4 are (1,4) and cells E4 and F4 are elements of (2,3,6). Since 8 cannot be in either of these groups, it must be in the 24x at right. (This can also be shown using divisibility rules.) If there is an 8 in the 24x clue, then it must be solved by (1,3,8).

In Step 2, we proved that the 2, 3, and 6 in Column I must solve cells in the 12x group or the 3/ group. Then cell I4 must be solved by the 8 and cell H4 by the 3.

Follow-up:

We see that 3 solves cell H4. Then cells E4 and F4 are (2,6), leaving another 3 for cell E3.

	A	B	C	D	E	F	G	H	I
1	?			7	2/		?		12x
2	4536x		42x	21+		84x		?	
3		7			3				1
4	7	9	28x		36x		5	3	8
5	?	1-	7	6-	?	14+			17+
6			3/		7				
7		?		36x	36x		?		3/
8								?	
9		?							7

Step 6: Cell C2 is a 1

Explanation:

In Column C, we have a 3/ clue that can be solved by (1,3), (2,6), or (3,9) and a 42x clue that can be solved by (1,6,7), or (2,3,7). Consider the possibilities for the 3/ clue. It is impossible for the 3/ group to be solved by (2,6) because 2 eliminates the (2,3,7) possibility for the 42x and 6 eliminates the (1,6,7) possibility. This is guaranteed because the 7 occupies cell B3, rather than a cell in Column C. Since the 3/ cannot be (2,6), it must be (1,3) or (3,9). Both these options guarantee that a 3 solves either cell C6 or cell C7. Either way, Column C's 3 will be in the 3/ group. Therefore, the 42x group cannot be solved by (2,3,7), so it must be solved by (1,6,7). With a 1 in cell I3, the 1 in the 42x clue must solve cell C2.

Follow-up:

With the 1 in cell C2, 6 must solve cell C3. Furthermore, the 28x clue must be solved with a 4 in cell C4 and a 1 in cell D4. The 1 in cell C2 also forces the 3/ group to be solved by (3,9).

	A	B	C	D	E	F	G	H	I
1	?			7	2/		?		12x
2	4536x		1	21+		84x		?	
3		7	6		3				1
4	7	9	4	1	36x		5	3	8
5	?	1-	7	6-	?	14+			17+
6			3/		7				
7		?		36x	36x		?		3/
8								?	
9		?							7

Step 7: Cell C8 is a 2

Explanation:

In Column C, we have found the positions of the 1, 4, 6, and 7 with certainty, and we know the 3/ group is solved by (3,9). Out of the remaining digits, 2, 5, 8, only the first is a factor of 36, so 2 must solve cell C8.

Follow-up:

The five options for a 36x group are (1,4,9), (1,6,6), (2,2,9), (2,3,6), or (3,3,4). We can reduce the options to (2,2,9) and (2,3,6) because we have guaranteed at least one 2.

	A	B	C	D	E	F	G	H	I
1	?			7	2/		?		12x
2	4536x		1	21+		84x		?	
3		7	6		3				1
4	7	9	4	1	36x		5	3	8
5	?	1-	7	6-	?	14+			17+
6			3/		7				
7		?		36x	36x		?		3/
8			2					?	
9		?							7

Step 8: Cell D9 is a 9

Explanation:

A 6- group can be solved by (1,7), (2,8), or (3,9). The 6- clue in Column D cannot be solved by (1,7) because both digits already appear in Column D. The 6- group also cannot be solved by (3,9). This is because the five options for a 36x group are (1,4,9), (1,6,6), (2,2,9), (2,3,6), and (3,3,4). Because a 2 solves cell C8, we can eliminate the choices down to (2,2,9) and (2,3,6). If the 6- clue was solved by (3,9), then both of these remaining 36x options would be eliminated. Therefore, the 6- clue is solved by (2,8). Since the 2 in Column D is used in the 6- clue, the 36x group must be solved by (2,3,6).

There are three remaining digits of which we do not know the relative positions in Column D: 4, 5, and 9. Notice that two of these numbers will solve the 21+ clue along with a 7. Suppose these two numbers are 5 and 9. This case is impossible because these three numbers would already add to 21, and there is still one more digit in cell E2 to be in the sum. Similarly, these two numbers cannot be 4 and 9 because this would force cell E2 to be a 1. However, the 1 in Column E is in the vertical 36x clue. Therefore, the two numbers must be 4 and 5. This leaves the 9 to solve cell D9.

Follow-up:

We see that 4 and 5 solve cells D2 and D3. Since the sum of 4,5, and 7 is 16, this forces cell E2 to be solved by a 5. With a 5 in cell E2, cell D2 cannot be a 5, so it must be a 4. Then cell D3 is a 5.

	A	B	C	D	E	F	G	H	I
1	?			7	2/		?		12x
2	4536x		1	4	5	84x		?	
3		7	6	5	3				1
4	7	9	4	1	36x		5	3	8
5	?	1-	7	6-	?	14+			17+
6			3/		7				
7		?		36x	36x		?		3/
8			2					?	
9		?		9					7

Step 9: Cell H2 is a 8

Explanation:

In Row 2, the 8 cannot be in the 12x or 84x groups because 8 is a factor of neither. After dividing 4536 by 7 and then by 9, we are left with 72. Although 8 is a factor of 72, it cannot be in the 4536x group either. If we have an 8 in the 4536x clue, the remaining two digits have to multiply to 72/8 or 9, giving the two possibilities of (1,8,9) and (3,3,8). The former is impossible because two 1s already appear in Rows 2 and 3, and the latter is impossible because this would force a 3 into cell A3 and there is already a 3 in cell E3. Therefore an 8 cannot appear in the 4536x clue. The only remaining option for the 8 in Row 2 is cell H2.

Follow-up:

Using similar logic for the 8 in Row 3, the 8 is limited to cell G3 or cell H3. With an 8 in cell H2, the 8 in Row 3 must solve cell G3.

We can also show the 9 in Row 3 solves cell H3 because it is not a factor of 84. If the 9 in Row 3 solved cell A3, then the product of cells A2 and B2 must be 8, giving two options: (1,8) and (2,4), both of which are impossible because a 1 and 8 already have been deduced in Row 2 in cells C2 and H2, respectively. Therefore, cell H3 is solved by a 9.

	A	B	C	D	E	F	G	H	I
1	?			7	2/		?		12x
2	4536x		1	4	5	84x		8	
3		7	6	5	3		8	9	1
4	7	9	4	1	36x		5	3	8
5	?	1-	7	6-	?	14+			17+
6			3/		7				
7		?		36x	36x		?		3/
8			2					?	
9		?		9					7

Step 10: Cell A3 is a 4

Explanation:

In Row 3 we have only the 2 and 4 remaining. If 2 solved cell A3, then the product of cells A2 and B2 must be 36, giving one possibility: (4,9). However, the 4 in Row 2 already solves cell D2, so it cannot be in the 4536x group and thus cell A3 cannot be 2. Therefore, cell A3 is a 4.

Follow-up:

With a 4 in cell A3, the 2 must solve cell F3. Then the product of cells F2 and G2 is 42, so they are 6 and 7 in some order. Also, 2 cannot solve cell F4 so 6 must solve it. Then, 2 must solve cell E4. Since 6 solves cell F4, 7 must solve cell F2, and 6 must solve cell G2.

Furthermore, because 4 solves cell A3, the product of cells A2 and B2 is 18. With a 6 in either cell F2 or cell G2, it cannot be in either cell A2 or cell B2. The only other option to multiply to 18 other than (3,6) is (2,9). The 9 in cell B4 forces the 9 to solve cell A2, leaving cell B2 to be solved by a 2.

The last digit in Row 2 is a 3, which goes in cell I2. This makes cell I1 to be solved by a 4.

	A	B	C	D	E	F	G	H	I
1	?			7	2/		?		4
2	9	2	1	4	5	7	6	8	3
3	4	7	6	5	3	2	8	9	1
4	7	9	4	1	2	6	5	3	8
5	?	1-	7	6-	?	14+			17+
6			3/		7				
7		?		36x	36x		?		3/
8			2					?	
9		?		9					7

Step 11: Cell E5 is a 8

Explanation:

In Column E, the 36x group is solved by (1,4,9). Then the digits 6 and 8 must be in cells E1 and E5. If 8 were to solve cell E1, then cell F1 needs to be solved by a 4. There is already a 4 in cell I1, so the 8 cannot be in cell E1. Therefore, the 8 in Column E solves cell E5.

Follow-up:

Clearly, the 6 in Column E has to solve cell E1 as it is the only choice left. Then cell F1 needs to be solved by a 3 because it needs to satisfy the 2/ clue given.

The 6- clue in Column D is solved by (2,8). Since 8 solves cell E5, then the 2 must solve cell D5, and the 8 solves cell D6.

	A	B	C	D	E	F	G	H	I
1	?			7	6	3	?		4
2	9	2	1	4	5	7	6	8	3
3	4	7	6	5	3	2	8	9	1
4	7	9	4	1	2	6	5	3	8
5	?	1-	7	2	8	14+			17+
6			3/	8	7				
7		?		36x	36x		?		3/
8			2					?	
9		?		9					7

Step 12: Cell H5 is a 4

Explanation:

In Column I, we are missing the 5 and 9, which are to solve cells I5 and I6. They are both in the 17+ group. Since 5 and 9 sum to 14, the other two digits must sum to 3. The only way to solve this is 1 and 2.

In the 14+ group, we have already deduced that cell G4 is solved by a 5. The other four cells must add to 9. In Column H, we already have solved for the relative positions of the 1, 2, and 3. The 1 and 2 will be in cells H6 and H7, and the 3 solves cell H4. The next smallest choice is a 4. If 4 solves cell H5, then the other three add to 5, which has two options: (1,1,3) and (1,2,2). Cell H5 can also be solved by a 5, which has one option: (1,1,2). This would force cell G5 to be solved by a 2, when there is already a 2 in cell D5. Therefore, cell H5 is not solved by a 5. If cell H5 were solved by a 6, the other three digits add to 3, which is impossible. Therefore cell H5 is solved by a 4.

Follow-up:

In fact, we know the other three digits in the 14+ cell are (1,1,3) because we cannot have the (1,2,2) option. This means 3 must solve cell G5, and 1 must solve both cells F5 and G6.

Another way to show that cell H5 cannot be solved by a 5, we can fill in some more numbers on the right. In the 3/ group in Column I, because the 3 solves cell I2, the 3/ must be solved by (2,6). With a 2 in cell C8, the 2 in the 3/ group must be in cell I7, and 6 must solve cell I8. In the 17+ clue, we know that cells H6 and H7 are solved by 1 and 2. Since there is a 2 in cell I7, the 2 in the 17+ must appear in cell H6, and then 1 must be in cell H7. The 2 in cell H6 will lead to the 2 in cell D5, and we can make the same argument as above.

Step 13: Cell I5 is a 9

Explanation:

In Row 5, there are only the 5, 6, and 9 not solved. There are 9s in cells A2 and B4, so the 9 cannot be in cell A5 or cell B5. Therefore, the 9 in Row 5 is forced to be in cell I5.

	A	B	C	D	E	F	G	H	I
1	?			7	6	3	?		4
2	9	2	1	4	5	7	6	8	3
3	4	7	6	5	3	2	8	9	1
4	7	9	4	1	2	6	5	3	8
5	?	1-	7	2	8	1	3	4	17+
6			3/	8	7		1	2	
7		?		36x	36x		?	1	2
8			2				?		6
9		?		9					7

Follow-up:

With the 9 in cell I5, cell I6 must be solved by a 5.

On an unrelated note, because a 6 solves cell I8, cell D8 cannot be solved by a 6. Therefore, the 3 must solve cell D8, and the 6 must solve cell D7.

We can also use the rule that one copy of each digit is allowed per row and per column to deduce that cell H1 is solved by a 5, cell G1 by a 9 (because there is a 9 in the 3/ group in Column C), cell A1 by a 2, cell B1 by a 1, and cell C1 by a 8.

In Column C, the only cell left for a 5 is in cell C9.

	A	B	C	D	E	F	G	H	I
1	2	1	8	7	6	3	9	5	4
2	9	2	1	4	5	7	6	8	3
3	4	7	6	5	3	2	8	9	1
4	7	9	4	1	2	6	5	3	8
5	?	1-	7	2	8	1	3	4	9
6			3/	8	7		1	2	5
7		?		6	36x		?	1	2
8			2	3				?	6
9		?	5	9					7

Step 14: Cell B5 is a 5

Explanation:

In Row 5, we are only missing the 5 and 6. Although we have no 5s or 6s to directly eliminate either choice, if we consider what combination of digits could solve the 1- clue, with 1, 2, 7, and 9 already occupying cells in Column B, the only choices left are (3,4), (4,5), and (5,6). If Cell B5 is solved by a 6, this would force a 5 to solve cell B6, but there is already a 5 in Row 6 in cell I6. This case is impossible, so cell B5 is not solved by a 6. Therefore, cell B5 is solved by a 5.

Follow-up:

Cell A5 is solved by a 6.

	A	B	C	D	E	F	G	H	I
1	2	1	8	7	6	3	9	5	4
2	9	2	1	4	5	7	6	8	3
3	4	7	6	5	3	2	8	9	1
4	7	9	4	1	2	6	5	3	8
5	6	5	7	2	8	1	3	4	9
6		1-	3/	8	7		1	2	5
7		?		6	36x		?	1	2
8			2	3				?	6
9		?	5	9					7

Step 15: Cell B6 is a 6

Explanation:

In Row 6, we are missing a 6. The only empty cell without a 6 in its column is cell B6.

Follow-up:

The following can be solved for by following the rule of one use of each digit per row and per column:

Cell H8 is solved by a 7, and cell H9 by a 6.

Cell G7 is also solved by a 7, cell G9 by a 2, and cell G8 by a 4.

Cell F6 is solved by a 4, cell C6 by a 9, cell A6 by a 3, and cell C7 by a 3.

Cell B9 is solved by a 3, cell B7 by a 4, and cell B8 by a 8.

Cell E9 is solved by a 4, cell E8 by a 1, cell E7 by a 9.

Cell F8 is solved by a 9, cell F7 by a 5, cell F9 by a 8.

Cell A7 is solved by an 8, cell A8 by a 5, and cell A9 by a 1.

And we arrive at the finished grid below:

	A	B	C	D	E	F	G	H	I
1	2	1	8	7	6	3	9	5	4
2	9	2	1	4	5	7	6	8	3
3	4	7	6	5	3	2	8	9	1
4	7	9	4	1	2	6	5	3	8
5	6	5	7	2	8	1	3	4	9
6	3	6	9	8	7	4	1	2	5
7	8	4	3	6	9	5	7	1	2
8	5	8	2	3	1	9	4	7	6
9	1	3	5	9	4	8	2	6	7